Fertiliser & Nutrient Management

Nutrients, soil testing, products, and responsible use for US growers

A detailed reference guide • United States
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Note: *This guide compiles general, widely accepted agronomic practice for US growers. Conditions vary by state, soil, and season — confirm specifics with your local Cooperative Extension office (find yours at nifa.usda.gov/land-grant-colleges-and-universities-partner-website-directory) and always follow current product labels and local regulations.*

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1. Essential Plant Nutrients

Plants require 17 essential elements. Carbon, hydrogen, and oxygen come from air and water; the remaining 14 come from the soil and are the ones fertiliser supplies. They are grouped by how much the plant needs — but all are essential, and a shortage of any one limits growth no matter how plentiful the others (the “law of the minimum”).

Group	Nutrients	Role (summary)
Primary macronutrients	Nitrogen (N), Phosphorus (P), Potassium (K)	Needed in largest amounts; the three numbers on fertiliser bags
Secondary macronutrients	Calcium (Ca), Magnesium (Mg), Sulfur (S)	Needed in moderate amounts; often supplied by lime, gypsum, or soil
Micronutrients	Fe, Mn, Zn, Cu, B, Mo, Cl, Ni	Needed in tiny amounts but still essential; deficiencies are crop- and soil-specific

What each primary nutrient does

- **Nitrogen (N)** drives leafy, vegetative growth and gives deep green color; it is the most commonly deficient and most mobile nutrient.
- **Phosphorus (P)** supports roots, energy transfer, flowering, and seed/fruit development; important early in growth.
- **Potassium (K)** regulates water use, strengthens stalks, and improves disease and stress tolerance and crop quality.

2. Understanding the N-P-K Label

Every fertiliser bag carries three numbers, such as 10-10-10 or 46-0-0. They are the guaranteed percentages by weight of nitrogen (N), available phosphate (P_2O_5), and soluble potash (K_2O), in that order.

- A 50 lb bag of 10-10-10 contains 5 lb each of N, P_2O_5, and K_2O (10% of 50).
- “Complete” fertilisers contain all three; “incomplete” supply one or two (e.g., urea 46-0-0 is N only).
- The ratio matters: a high-first-number product (e.g., 46-0-0) is for nitrogen; a balanced 10-10-10 is a general feed; a high-middle product aids establishment and rooting.
- To compare products, compute the cost per pound of actual nutrient, not per bag.

Note: *Phosphate and potash are expressed as oxides (P_2O_5 and K_2O) by convention, not as elemental P and K. Soil test recommendations in the US are usually given in these same oxide units, so the two line up directly.*

3. Soil Testing: The Starting Point

Fertilising without a soil test is guessing — it wastes money, risks crop damage from over-application, and pollutes water. A soil test is inexpensive and is the single most valuable input decision a grower

makes.

How to sample

- Take 10–15 cores from across a uniform area, to plow depth (about 6–8 in for cropland), and mix into one composite sample.
- Sample separate areas (different soil types, slopes, or management) separately.
- Avoid odd spots — old manure piles, fence lines, wet holes.
- Test every 2–3 years, at the same time of year for comparable trends (fall is common).
- Use a reputable lab (often through your Extension service) and state the crop you plan to grow so recommendations are tailored.

What the report tells you

- Soil pH and lime requirement.
- Levels of P, K, and usually Ca, Mg, and sometimes micronutrients and organic matter.
- Fertiliser and lime recommendations in lb/acre (or per 1,000 sq ft) for your target crop and yield.

4. Soil pH & Liming

Soil pH controls how available nutrients are to plants. Even with ample fertiliser, nutrients can be locked up if pH is wrong. Most crops do best at pH 6.0–7.0, where the broadest range of nutrients is available.

- **Too acidic (low pH)** ties up phosphorus and can release toxic aluminum and manganese; correct by applying agricultural lime (calcitic or, where Mg is low, dolomitic).
- **Too alkaline (high pH)** locks up iron, manganese, zinc, and phosphorus; lower with elemental sulfur or acid-forming fertilisers.
- Lime acts slowly — apply months ahead (ideally the fall before) and incorporate; retest before re-liming.
- A few crops prefer acidity (blueberries ~4.5–5.5, potatoes slightly acidic to limit scab) — match pH to the crop.

Note: *Always base lime rates on the soil test's lime-requirement (buffer pH) value, not on pH alone — two soils at the same pH can need very different amounts of lime depending on their buffering capacity.*

5. Types of Fertiliser

Synthetic (mineral) fertilisers

Product	Analysis	Notes
Urea	46-0-0	Most concentrated dry N; incorporate or it can volatilize
Ammonium nitrate	34-0-0	Quick N; less common now due to regulation

Product	Analysis	Notes
UAN solution	28-0-0 / 32-0-0	Liquid N for spraying/injection
Anhydrous ammonia	82-0-0	Cheapest N per lb; injected; requires care/training
MAP	11-52-0	High phosphate starter
DAP	18-46-0	Common phosphate source
Potash (MOP)	0-0-60	Standard potassium source
Balanced blends	10-10-10, 19-19-19	General-purpose complete fertilisers

Slow-release & stabilized

Coated or stabilized products (polymer-coated urea, products with nitrification/urease inhibitors) release nitrogen gradually, reducing losses and feeding the crop more steadily — useful on sandy soils and for reducing leaching.

Organic fertilisers

Derived from plant, animal, or mineral sources (manure, compost, blood meal, bone meal, feather meal, fish emulsion, greensand). They release nutrients more slowly as soil organisms break them down, improve soil structure and biology, and are lower-analysis, so larger amounts are needed per unit of nutrient.

6. Organic Sources: Manure & Compost

Animal manure and compost recycle nutrients, add organic matter, and improve soil structure and water-holding capacity. Their nutrient content varies widely by source, bedding, and handling, so test before relying on them for precise rates.

Source	Relative nutrients	Notes
Poultry litter	High N & P	Potent; easy to over-apply P — test and rate carefully
Cattle / dairy manure	Moderate, balanced	Good all-round soil builder; bulky
Swine manure	High N (liquid)	Often injected; strong N source
Compost	Low, slow-release	Best as a soil conditioner and organic-matter builder

- Apply and incorporate well ahead of planting; raw manure can burn plants and carries food-safety concerns on edible crops (observe required intervals before harvest).
- Account for manure nutrients in your fertiliser plan so you do not double-apply, especially phosphorus, which builds up and runs off.
- Compost fully before use to kill weed seeds and pathogens and to stabilize nutrients.

7. Application Methods & Timing

Methods

Method	Description	Best for
Broadcast	Spread evenly over the surface, incorporated or not	Pre-plant fertility, lime, potash, whole-field needs
Banding / starter	Placed in a band near the seed at planting	Phosphorus and starter N for early vigor (keep off the seed)
Side-dress / top-dress	Applied beside or over an established crop	In-season N for corn (V6), top-dress N for wheat
Fertigation	Dissolved and delivered through irrigation	Vegetables, orchards, precise in-season feeding
Foliar	Sprayed on leaves for fast uptake	Correcting micronutrient deficiencies; supplemental, not primary

Timing

- **Split nitrogen** into two or more applications (some at planting, more at peak demand) to match crop uptake and cut leaching losses — especially on sandy soils.
- **Phosphorus and potassium** are less mobile and are commonly applied before or at planting.
- Avoid applying nitrogen far ahead of need or before heavy rain — much can be lost.
- Feed to the crop's demand curve: most crops take up the bulk of their nutrients during rapid vegetative growth and reproduction.

Note: *Keep fertiliser, especially nitrogen in direct seed contact, within safe limits — too much salt near germinating seed causes "fertiliser burn." Follow starter-fertiliser safe-rate guidelines for your crop and row spacing.*

8. Nutrient Needs by Crop

These are general guideposts. Actual rates depend on your soil test, yield goal, previous crop, and manure history — always start from a soil-test recommendation.

Crop	Nitrogen appetite	Notes
Corn	High	~0.8–1.2 lb N per bushel of yield goal; credit N from previous legume
Wheat	Moderate–high	Split: some at planting, top-dress in spring
Soybean & other legumes	Low (fixes own N)	Inoculate seed; focus on P, K, and S
Cotton	Moderate	Balance N to avoid excess vegetative growth
Leafy vegetables	High N	Steady N for continuous leaf growth
Fruiting vegetables	Balanced	Ease off heavy N at fruiting or you get leaves, not fruit
Root crops	Lower N, higher K	Excess N causes forking/leafy tops

9. Diagnosing Deficiencies

Nutrient deficiencies show characteristic patterns. A key clue is whether symptoms appear first on **old (lower) leaves** or **new (upper) leaves**: mobile nutrients are pulled from old growth to new, so their shortage shows on old leaves first; immobile nutrients show on new growth first.

Nutrient	Mobility / where symptoms show	Classic symptom
Nitrogen (N)	Mobile – older leaves first	Uniform yellowing (chlorosis), starting at leaf tips; stunting
Phosphorus (P)	Mobile – older leaves first	Dark green or purplish/reddish leaves; slow growth
Potassium (K)	Mobile – older leaves first	Yellowing/scorching along leaf margins
Magnesium (Mg)	Mobile – older leaves first	Interveinal yellowing (veins stay green)
Sulfur (S)	Immobile – new leaves first	General yellowing of young leaves
Iron (Fe)	Immobile – new leaves first	Interveinal yellowing on youngest leaves (common at high pH)
Calcium (Ca)	Immobile – new growth	Distorted new growth; blossom-end rot in tomato/pepper

Note: *Look-alikes abound: drought, waterlogging, root damage, herbicide injury, and disease can mimic deficiencies. Confirm suspicions with a soil test and, where needed, plant-tissue analysis before investing in corrective treatment.*

10. Environmental Stewardship (4R Nutrient Stewardship)

Fertiliser that leaves the field is money lost and a pollution problem — nitrogen leaching into groundwater and phosphorus running into lakes and rivers feed harmful algal blooms. The widely taught framework for responsible use is the “4Rs.”

The 4 Rs	Meaning
Right Source	Match the fertiliser type to crop need and soil conditions
Right Rate	Apply the amount the soil test and yield goal justify — no more
Right Time	Apply when the crop can use it, to limit loss
Right Place	Place nutrients where roots can reach them and runoff is minimized

- Keep phosphorus out of water: maintain buffer strips, avoid applying before heavy rain or on frozen ground, and incorporate where possible.
- Use cover crops to catch leftover nitrogen and hold soil over winter.
- Credit nutrients from manure and previous legume crops to avoid over-applying.
- Keep records of soil tests and applications — good for the crop, the budget, and compliance.

Note: *Rates, restrictions, and nutrient-management regulations vary by state and watershed. Base your plan on a current soil test and your local Cooperative Extension or certified crop adviser's recommendations.*